

Want:

Suppose L, K are compactly generated elementary groups and $K \curvearrowright X$ with comp/open point stabilizers (no need to be a faithful action). Let $U \in \mathcal{U}(L)$, then we have that

$$\gamma + \alpha \leq \mathcal{G}(LS_U(K, X))$$

$$\text{where } \mathcal{G}(L) = \gamma + 2, \mathcal{G}(K) = \alpha + 2$$

↳ First ideas:

1. Case K compact follows from the same argument as the other lemma

2. For the case K non-compact we will make an induction on γ and α , one after the other

3. Base case: $\gamma = 0$

$$\alpha < \mathcal{G}(K) \leq \mathcal{G}(LS_U(K, X))$$

4. Suppose that for all $\mathcal{Q}$ compactly generated non-trivial elementary group with $\mathcal{G}(\mathcal{Q}) < \gamma$ and for all $E \curvearrowright X$ elementary group with comp/op. pts stabilizers we have, for $U \in \mathcal{U}(\mathcal{Q})$

$$\delta + \beta < \mathcal{G}(\mathcal{Q}S_U(E, X))$$

$$\text{where } \mathcal{G}(\mathcal{Q}) = \delta + 2, \mathcal{G}(E) = \beta + 2$$

↑ (Induction hypothesis)

5 $\rightarrow$ Let L be s.t. $\mathcal{G}(L) = \delta + 2$ and K elementary with and compactly generated with $\mathcal{G}(K) = d + 2$

6 $\rightarrow$ Because K is non-compact, let $Z \subset X$ be an infinite orbit

7 $\rightarrow$ Let $U \in \mathcal{U}(L)$ and define

$$N := \langle \langle U \rangle \rangle_L, H := N S_0(K, Z), J := \bigoplus_Z (N, U)$$

8 $\rightarrow$ H is closed in $L S_0(K, X)$, then

$$\mathcal{G}(H) \leq \mathcal{G}(L S_0(K, X))$$

9 $\rightarrow$ Because $N \trianglelefteq L$ we have $\text{Res}(L) \leq N$ and, because L is compactly generated

$$\delta + 1 = \mathcal{G}(\text{Res}(L)) \leq \mathcal{G}(N) \leq \mathcal{G}(L) = \delta + 2$$

10 $\rightarrow$ Let $(Q_i)_{i \in \mathbb{N}}$ be an increasing family of compactly generated groups with $0 \leq Q_i \leq Q_{i+1}$ and $\bigcup_{i \in \mathbb{N}} Q_i = N$. Then

$$\sup_{i \in \mathbb{N}} \mathcal{G}(Q_i) \leq \delta + 1 \rightarrow \boxed{\text{This might not be true when } \mathcal{G}(N) = \delta + 2, \text{ so we cannot apply induction}}$$

11 $\rightarrow$ Define $H_i := Q_i S_0(K, Z)$ and $\mathcal{G}(Q_i) = \beta_i + 2$. By the induction hypothesis we have ~~$\beta_i + 2$~~ $\beta_i + 2 \leq \mathcal{G}(H_i)$

13 $\Rightarrow$ Then

$$\sup B_i + \alpha \leq \sup \mathcal{G}(H_i) \leq \sup \mathcal{G}(H)$$

$$\Rightarrow \gamma + \alpha \leq \sup(H) \leq \mathcal{G}(LS_0(K, X))$$

as desired.

~~Proof the new~~

The biggest problem I ran on this proof was the fact that we are not sure about the rank of N , so it is not possible to apply the induction

Also, at first I was trying to work with an idea similar to the other proof but I ended up not working with the $\text{Res}(H)$, which might be a problem

~~It~~ It might be necessary to add some steps in between and maybe work inside $\text{Res}(L)$ ~~instead~~ instead of N